



RISKS-X

风险分析预测与管控研究院



前沿与交叉科学研究院

SUSTech Academy for
Advanced Interdisciplinary Studies



SUSTech

Davide Zaccagnino, Jiawei Li & Didier Sornette

BEYOND COMPLETENESS

The Persistent Triggering Power of Earthquakes
at Unobservable Small Magnitudes

Institute of Risk Analysis, Prediction and Management (Risks-X)

Academy for Advanced Interdisciplinary Studies

Southern University of Science and Technology (SUSTech), Shenzhen, China

Haicheng Symposium HCEQ2025 – Shenyang, Liaoning; July 10, 2025

Davide Zaccagnino

Post-doctoral student

zaccagnino@sustech.edu.cn

Risks-X Institute of Risk Analysis, Prediction and Management
Academy for Advanced Interdisciplinary Studies

SUSTech Southern University of Science and Technology

Working on

✓ **Modeling and forecasting large earthquakes** with

- Statistical physics
- Stochastic processes (memory)
- Geodesy (strain localization)

✓ **Collective behaviors in fault systems**

✓ **Earthquake Physics** (nucleation)

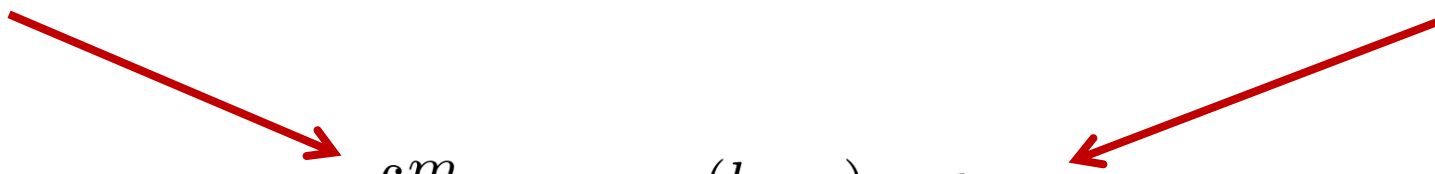


The ETAS model faces a fundamental conceptual and practical challenge known as the **ultraviolet (UV) catastrophe**.

This problem arises from the empirical observation that the Gutenberg-Richter b-value typically equals or exceeds the ETAS productivity exponent, implying that small earthquakes collectively dominate the triggering of subsequent events.

$$N(\geq m) = 10^{a-b(m-m_c)}$$

$$F_{m_0}(m) = K e^{\alpha(m-m_0)}$$


$$\int_{m_{\min}}^{m_{\max}} 10^{-(b-\alpha)m} dm$$

In the absence of a **lower magnitude cut-off**, and under standard ETAS assumptions, this leads to **a divergence in the total number of earthquakes at all magnitude values as $m_{\min} \rightarrow -\infty$** .

Specifically, the UV catastrophe emerges because smaller earthquakes, while individually producing weaker stress perturbations and few aftershocks, are exponentially more numerous as a function of their magnitudes, as described by the Gutenberg-Richter law.

The standard ETAS model

$$\lambda(x, y, t, m | \mathcal{H}_t) = \left[\mu(x, y) + \sum_{i|t > t_i} g(x - x_i, y - y_i, t - t_i, m_i - m_0) \right] \beta \exp(-\beta(m - m_0))$$

$$g(x - x_i, y - y_i, t - t_i, m_i) = K \exp(\alpha(m_i - m_0)) \frac{T_{norm} \exp\left(-\frac{t - t_i}{\tau}\right)}{(t - t_i + c)^p} \frac{S_{norm}}{[(x - x_i)^2 + (y - y_i)^2 + d \exp(\gamma(m_i - m_0))]^{1+\rho}}$$

addresses this UV divergence by assuming the existence of a minimum triggering magnitude

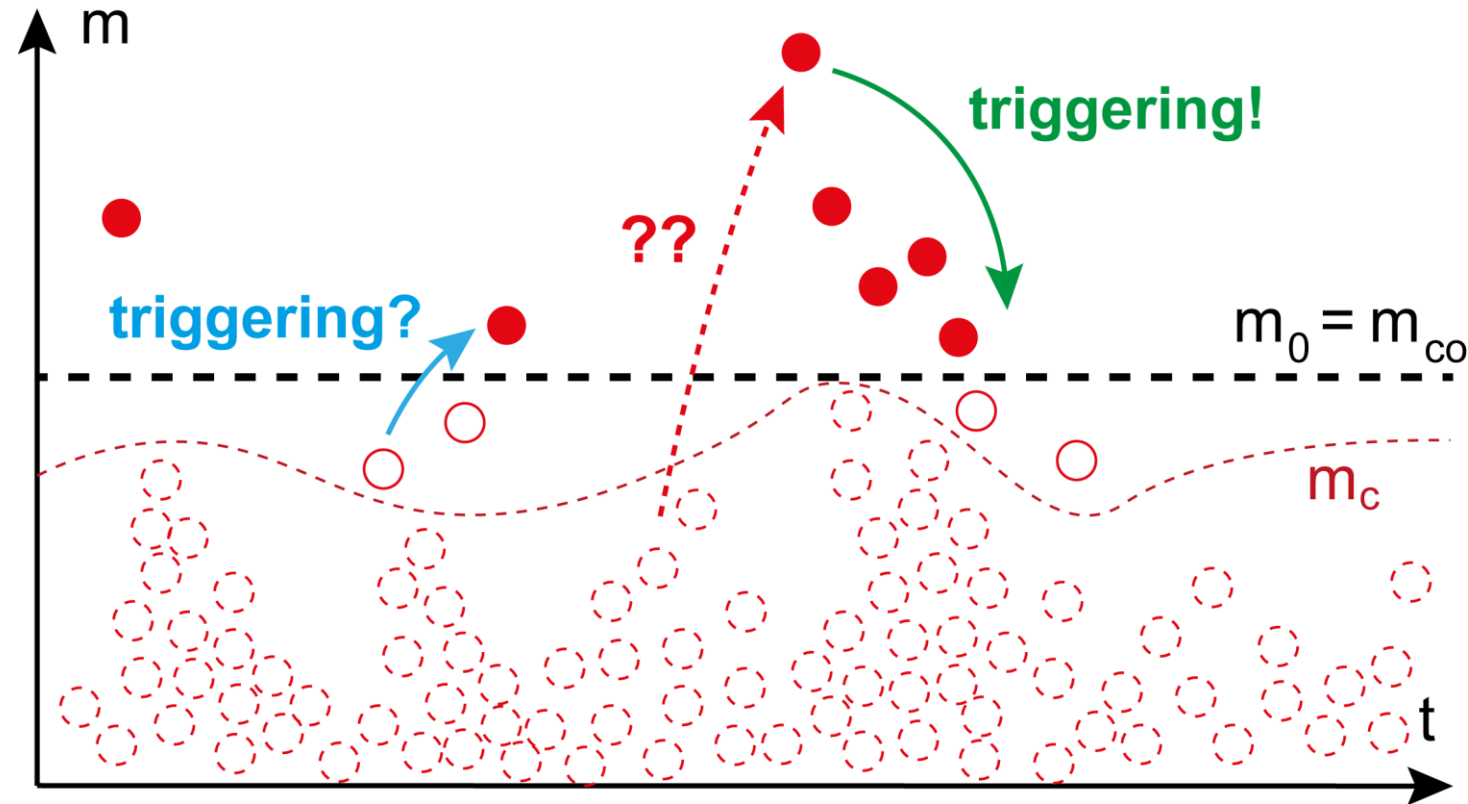
$$m_{min} := m_0$$

below which earthquakes cannot trigger aftershocks.

This magnitude is usually **arbitrarily** posed equal to the catalog cut-off magnitude

$$m_0 = m_{co} \geq m_c$$

$m_0 = m_{c0}$ is a guess and implies important consequences in terms of reliability of ETAS forecasts and estimations of its parameters (e.g., the branching ratio)



Therefore, it is important to measure m_0 reliably.

How can we measure m_0 reliably?

The Fertility-Tapered ETAS model

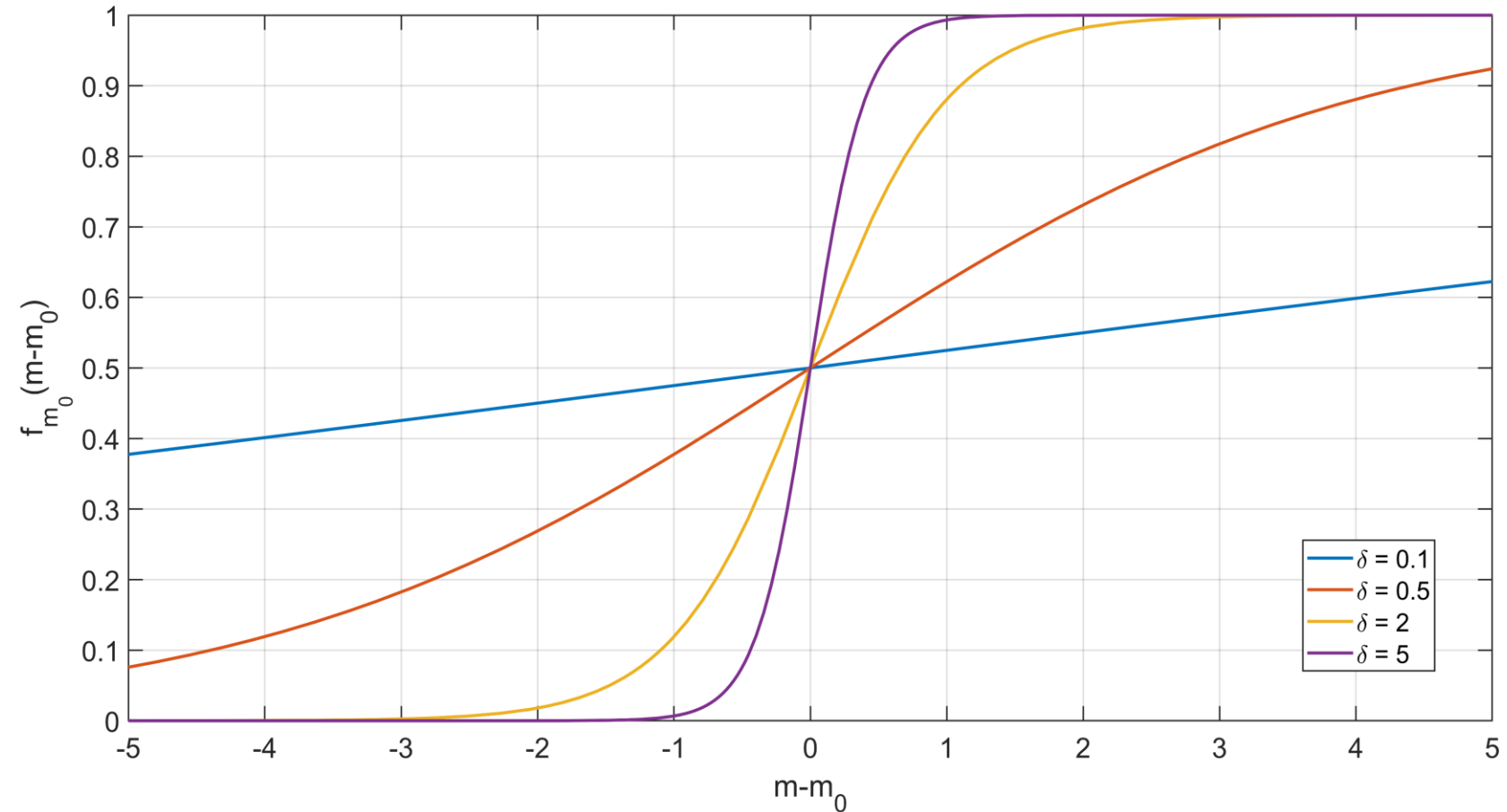


$$K e^{\alpha(m-m_0)} \longrightarrow K e^{\alpha(m-m_0)} \cdot f_{m_0}(m)$$

$1/\delta$

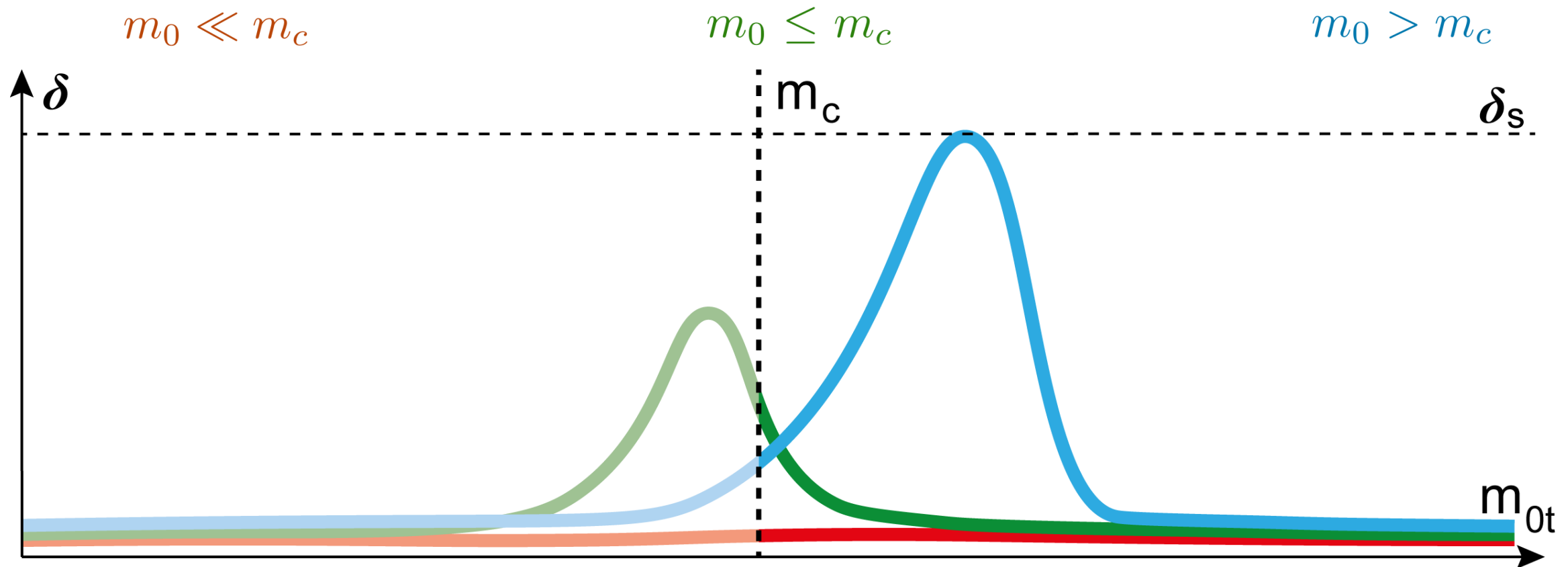


$$f_{m_0}(m) = \frac{e^{\delta(m-m_0)}}{1 + e^{\delta(m-m_0)}}$$

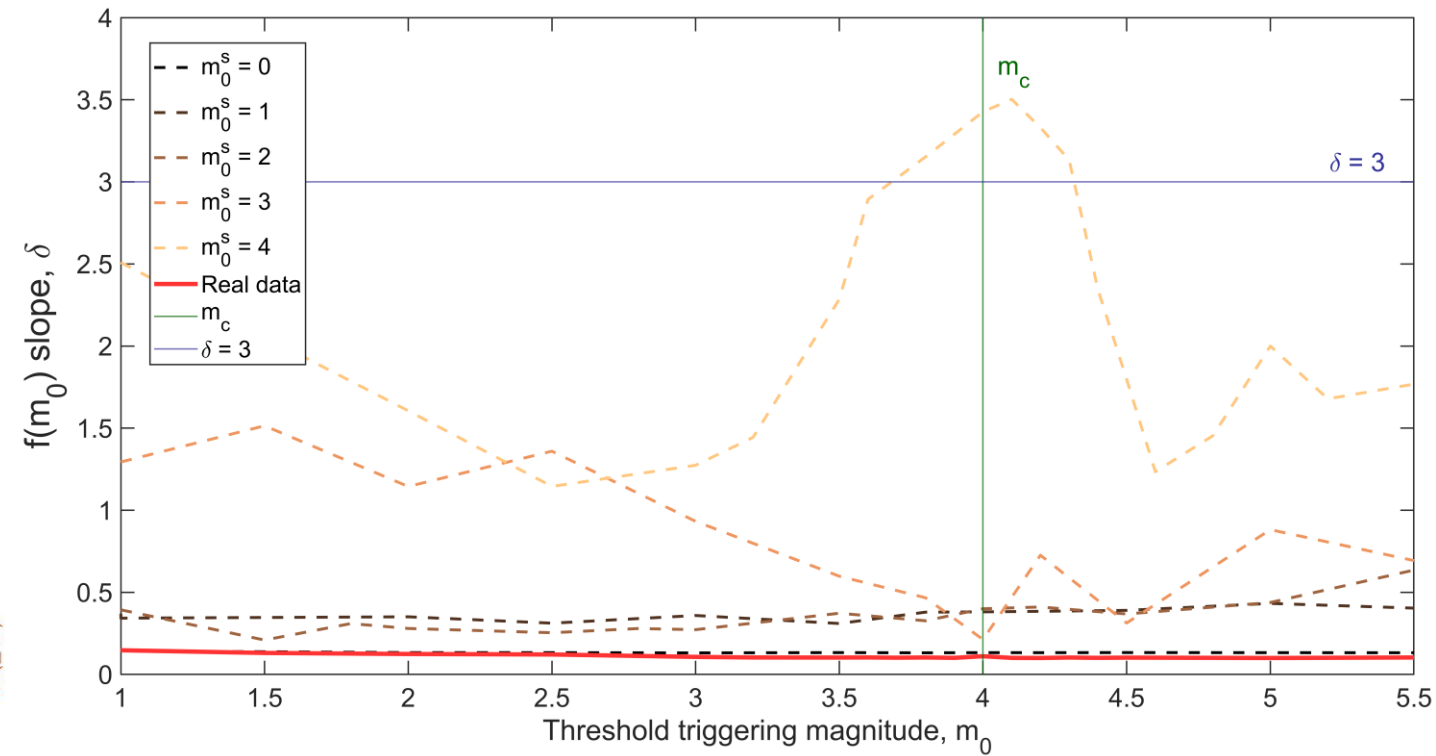
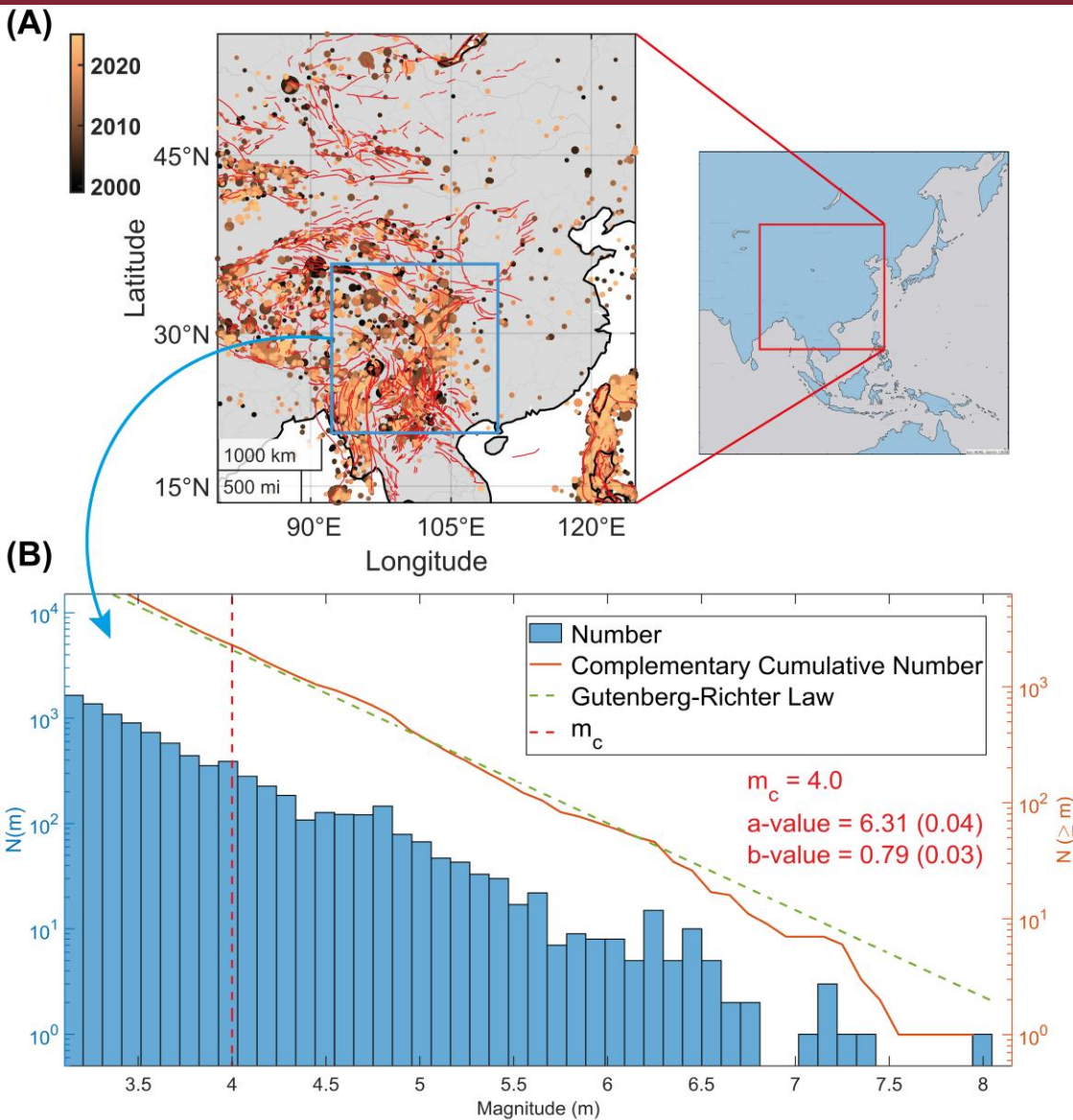


We perform the calibrations of simulated and real seismic catalogs investigating how δ changes as a function of a wide range of possible values of m_0 .

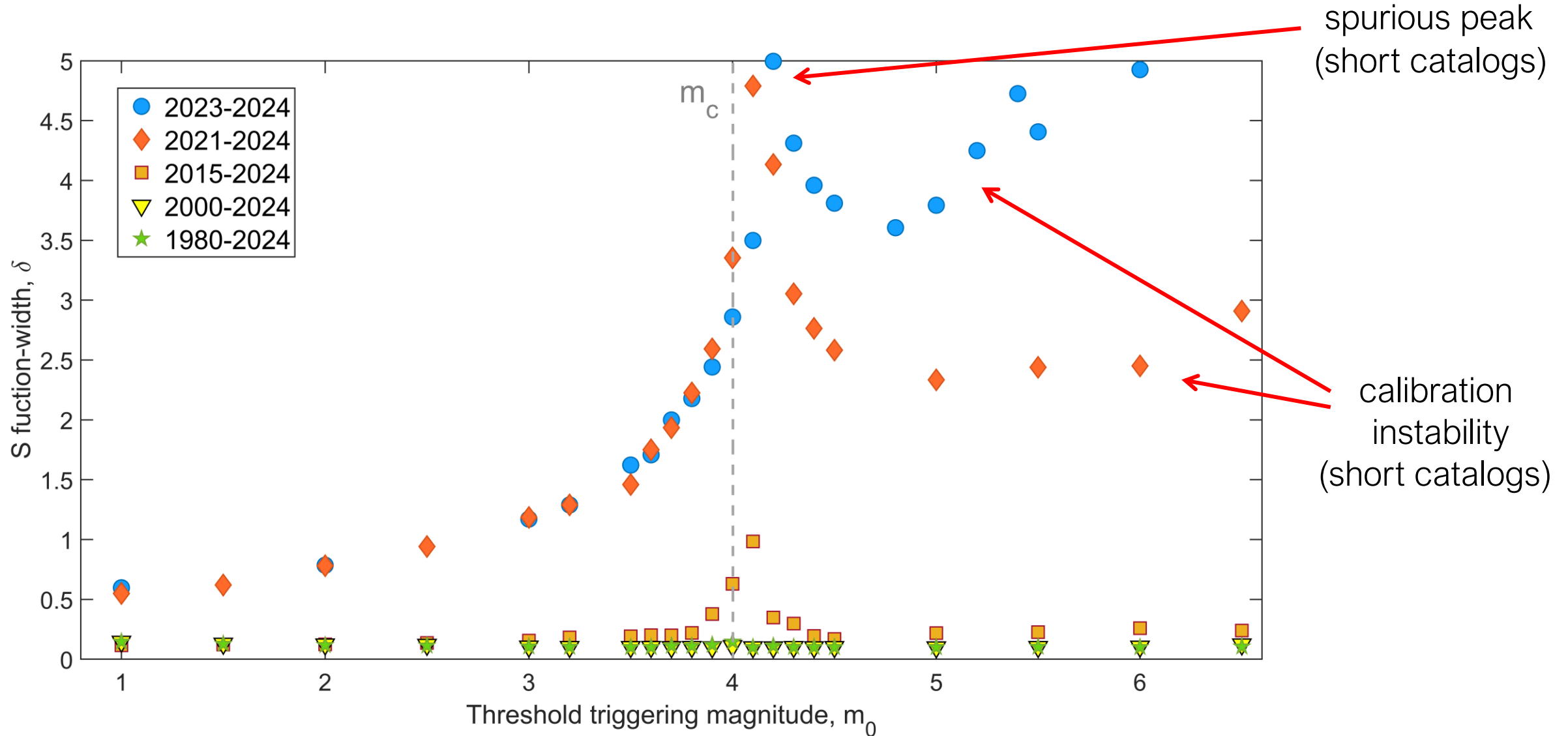
Three scenarios are possible:



Result for the CSES Region in China: $m_0 < m_c$



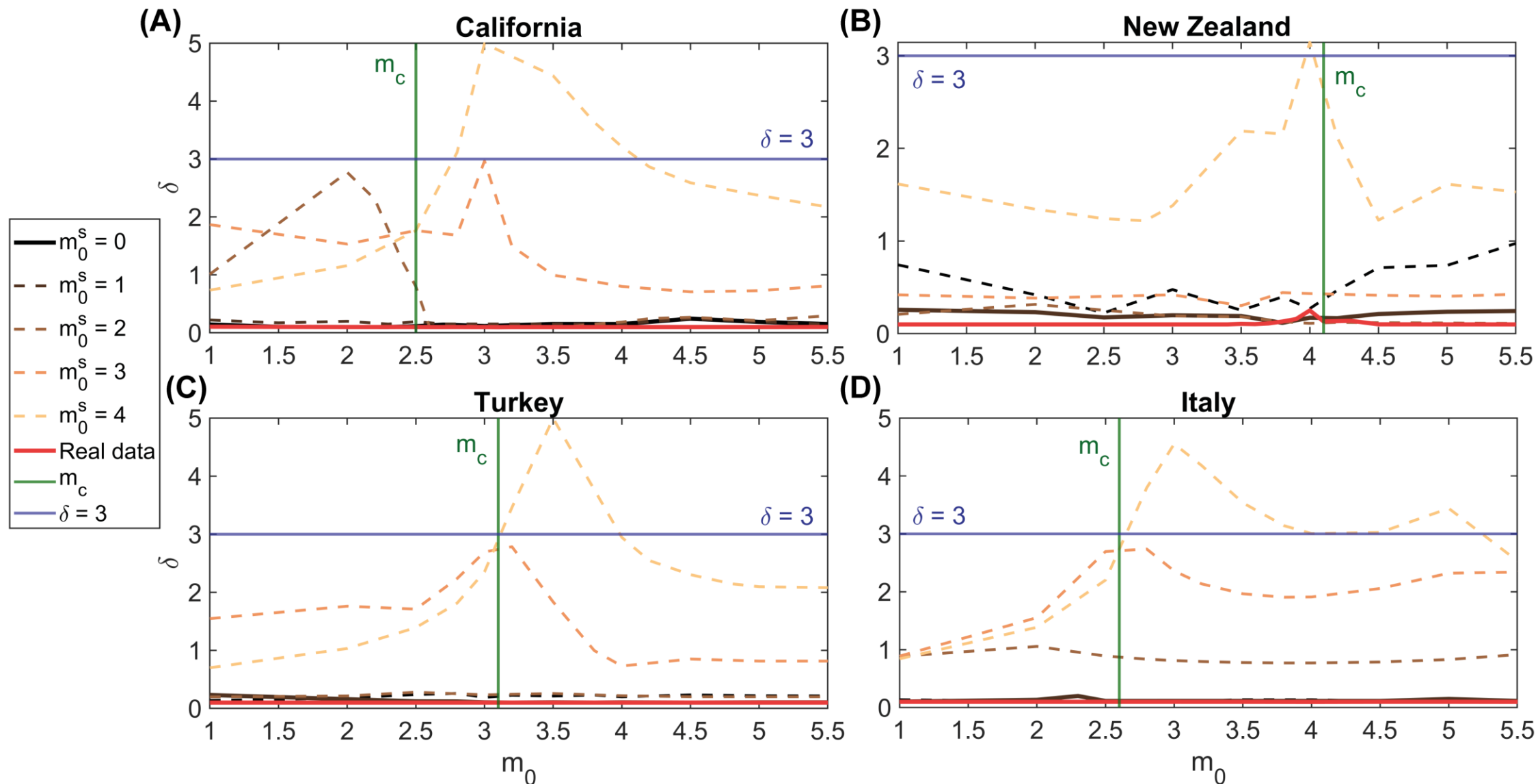
The Effect of Catalog Length and Completeness

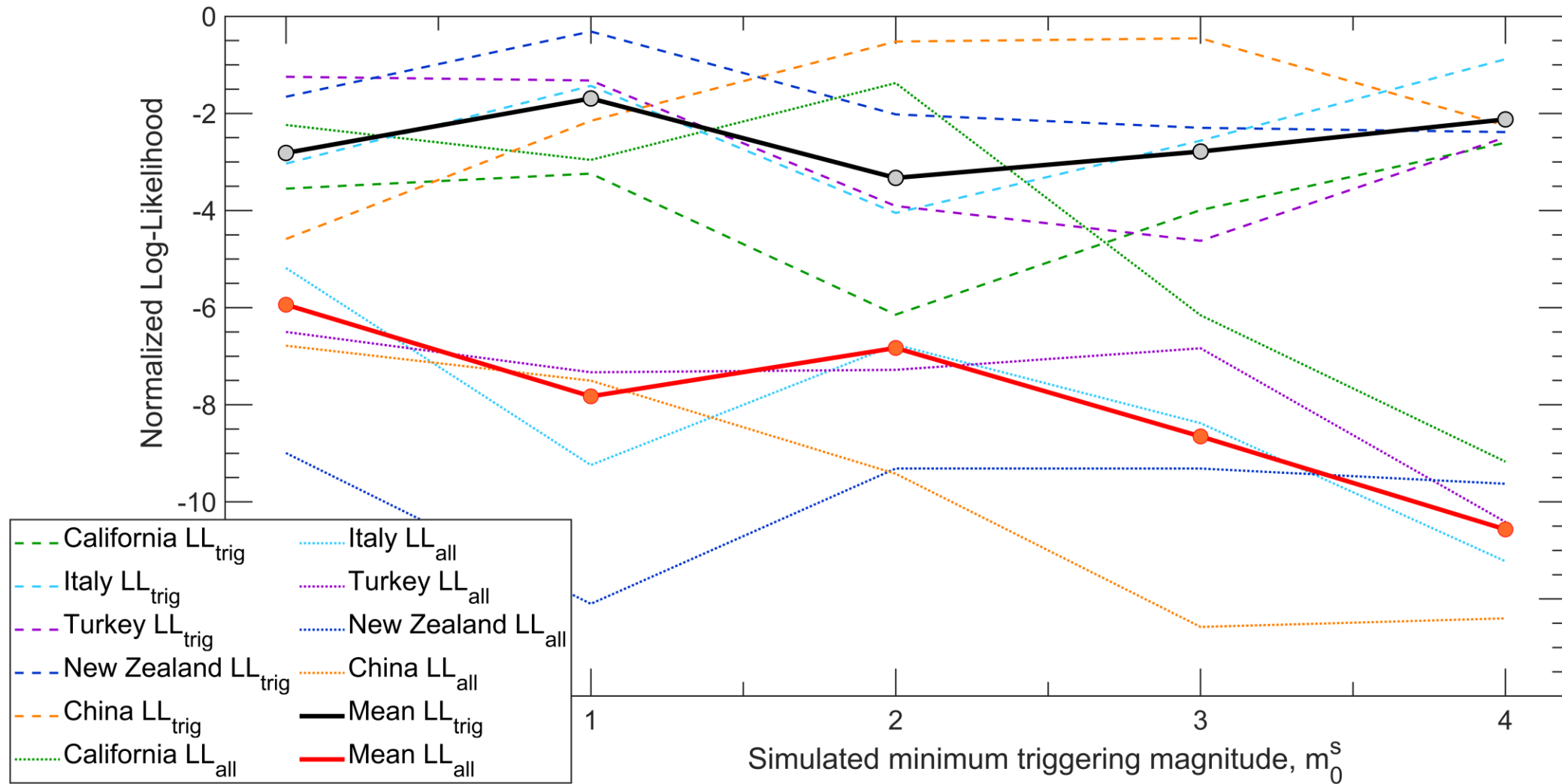


Results for Regional Catalogs Worldwide: $m_0 < m_c$



RISKS-X
风险分析预测与管控研究院





The standard ETAS model is nested within our proposed fertility-tapered ETAS model.

For nested models, the Wilks likelihood ratio test is ideal to assess whether, given the data, the more complex model is needed or if the simpler model is sufficient, and the more complex model can be rejected.

The test statistic is defined as

$$D = 2 (\log \mathcal{L}_1 - \log \mathcal{L}_0) = 2 (LL_1 - LL_0)$$

Under the null hypothesis (i.e., H_0 : standard ETAS is the true generating mechanism), the test statistic asymptotically follows a chi-squared distribution:

$$D \sim \chi_k^2$$

The p-value is calculated as

$$p = P(\chi_1^2 > D) = 1 - F_{\chi_1^2}(D)$$

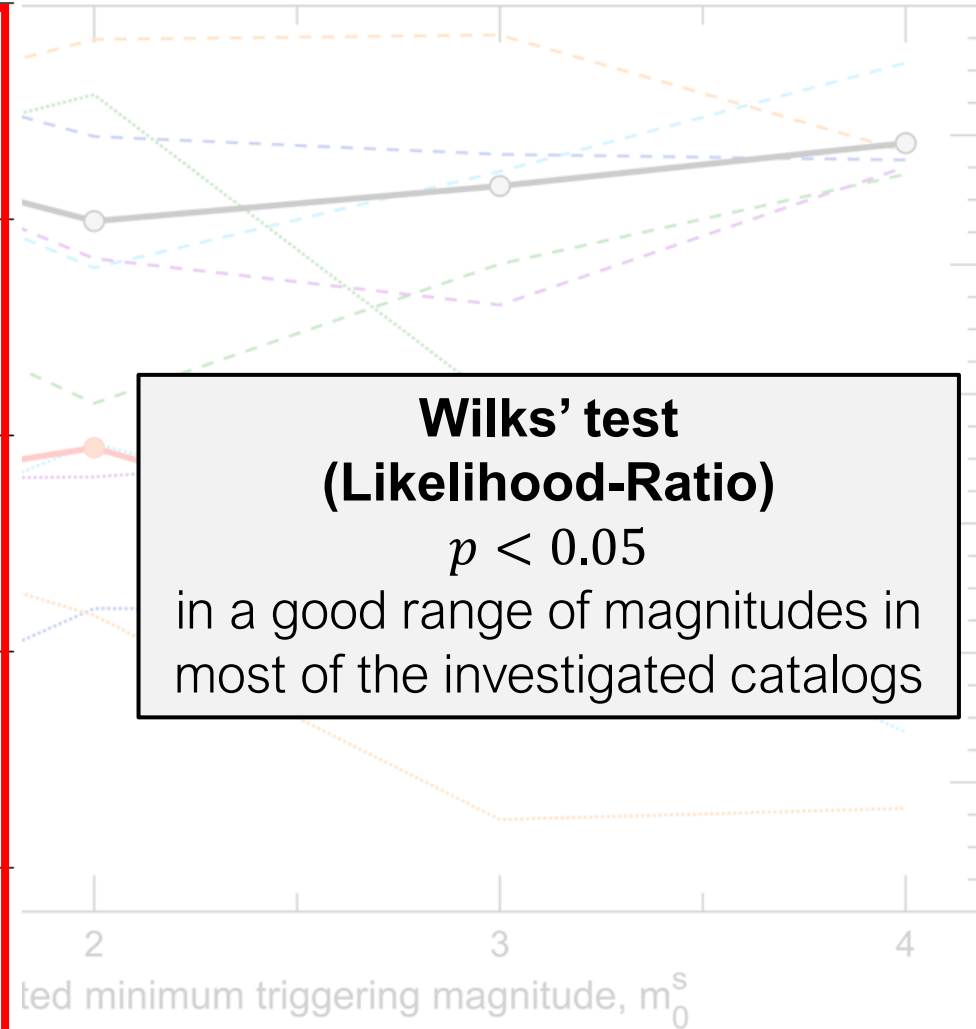
We consider the improvement statistically significant if $p < 0.05$.

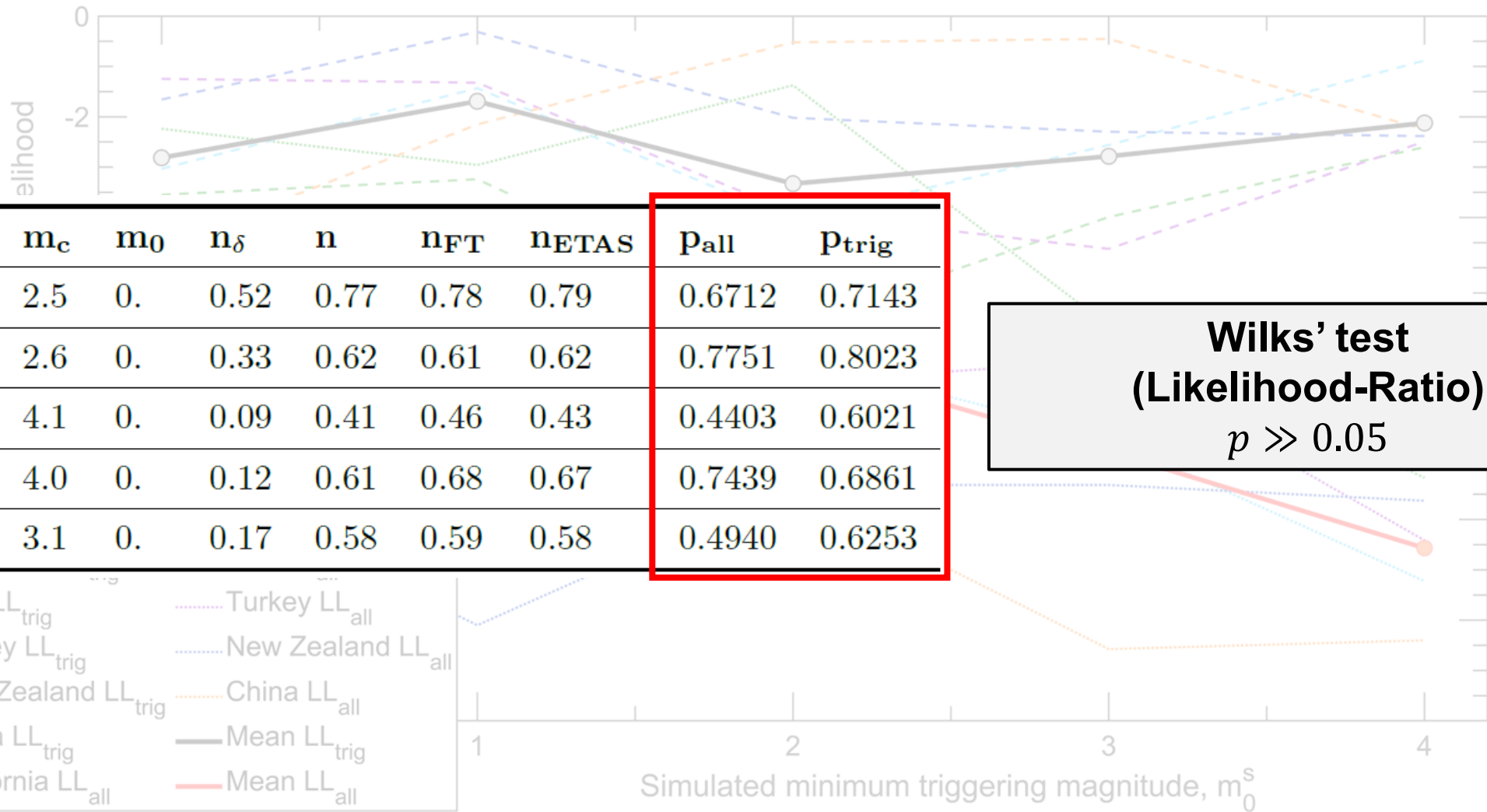
Test Outputs (Simulated Catalogs)



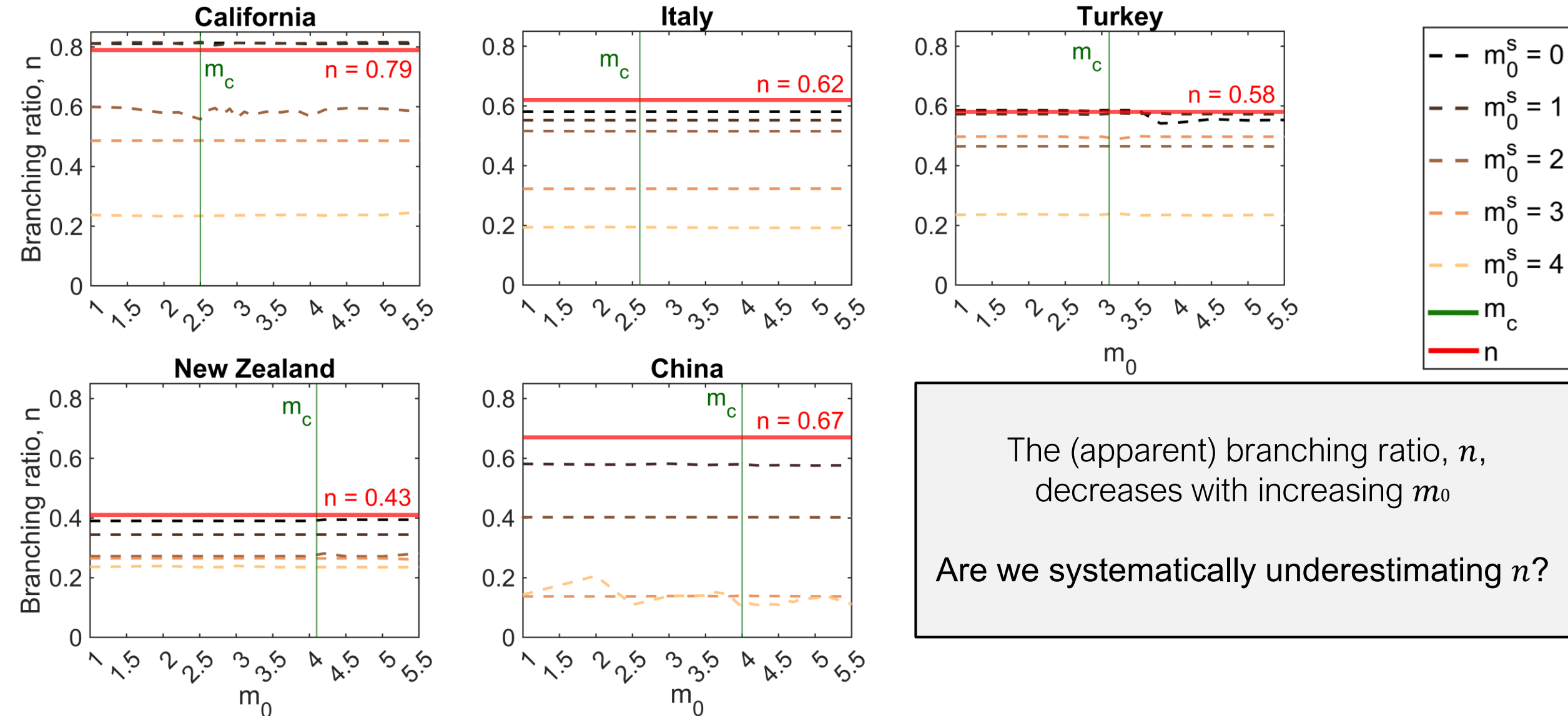
Catalog	m_0^s	m_c	LL_{all}	LL_{trig}	P_{all}	P_{trig}
California	0	2.5	-3.55	-2.24	0.0616	0.4841
California	1	2.5	-3.24	-2.96	0.0425*	0.3296
California	2	2.5	-6.14	-1.37	0.1942	0.1354
California	3	2.5	-3.99	-6.16	0.1059	0.0067*
California	4	2.5	-2.60	-9.17	0.0202*	0.0003*
Italy	0	2.6	-3.03	-5.18	0.0996	0.2935
Italy	1	2.6	-1.43	-9.24	0.0150*	0.0024*
Italy	2	2.6	-4.05	-6.76	0.4124	0.0389*
Italy	3	2.6	-2.56	-8.38	0.0560	0.0062*
Italy	4	2.6	-0.88	-11.22	0.0081*	0.0003*
New Zealand	0	4.1	-1.66	-8.99	0.0225*	0.0811
New Zealand	1	4.1	-0.31	-12.10	0.0049*	0.0023*
New Zealand	2	4.1	-2.02	-9.31	0.0342*	0.0550
New Zealand	3	4.1	-2.30	-9.31	0.0476*	0.0550
New Zealand	4	4.1	-2.38	-9.63	0.0523	0.0376*
China	0	4.0	-4.59	-6.78	0.0121*	0.0221*
China	1	4.0	-2.15	-7.50	0.0008*	0.0098*
China	2	4.0	-0.52	-9.42	0.0001*	0.0012*
China	3	4.0	-0.45	-12.58	0.0001*	0.0000*
China	4	4.0	-2.25	-12.40	0.0009*	0.0000*
Türkiye	0	3.1	-1.25	-6.50	0.0042*	0.1840
Türkiye	1	3.1	-1.32	-7.33	0.0045*	0.0642
Türkiye	2	3.1	-3.91	-7.28	0.0890	0.0682
Türkiye	3	3.1	-4.62	-6.84	0.2250	0.1179
Türkiye	4	3.1	-2.49	-10.41	0.0167*	0.0020*

--- Cali
 --- Italy
 --- Tur
 --- Nev
 --- Chir
 --- Cali





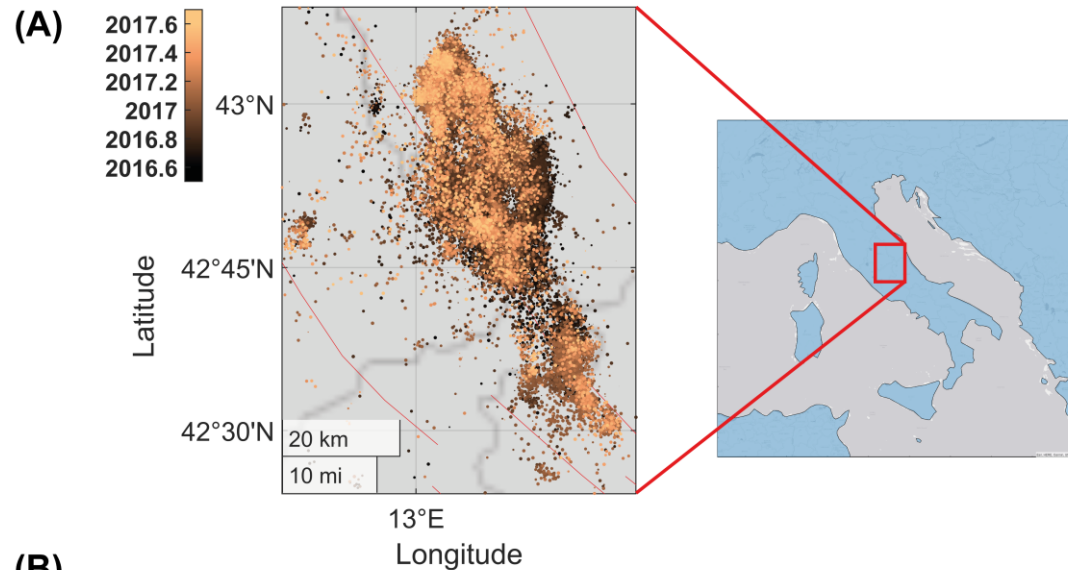
Arbitrary m_0 Biases ETAS Parameter Estimation



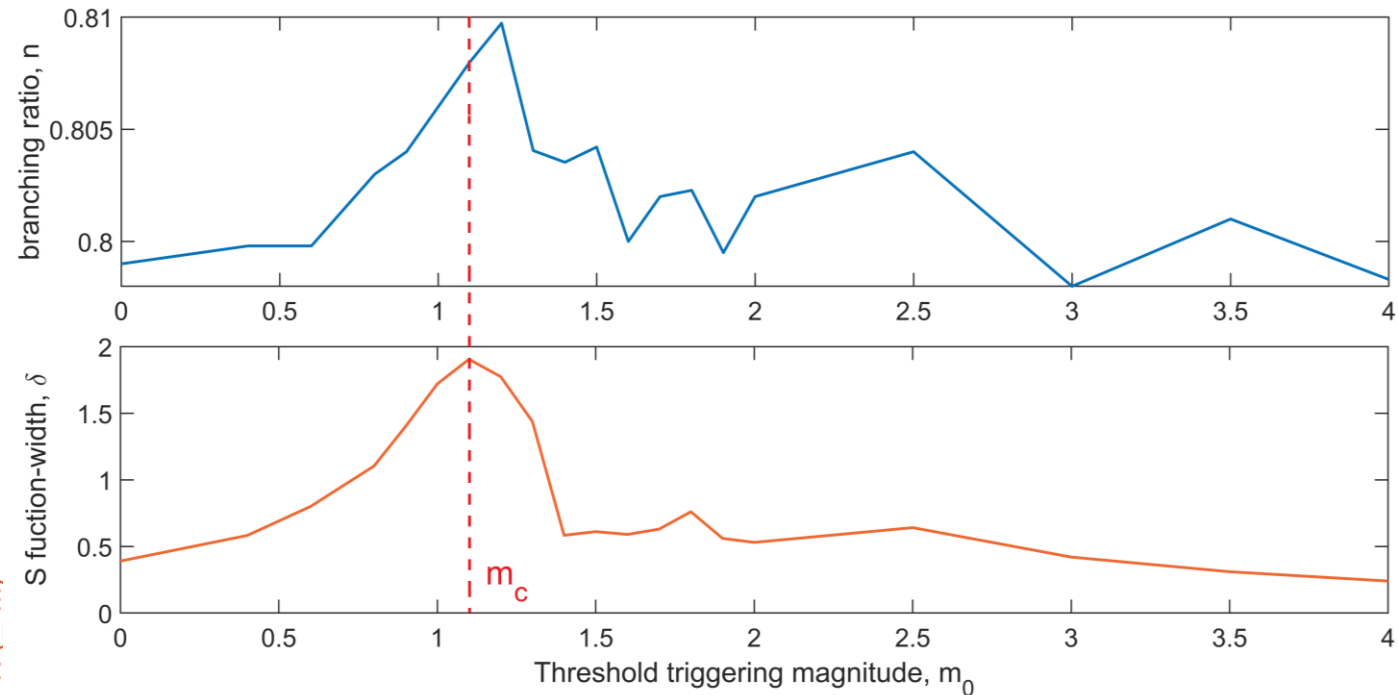
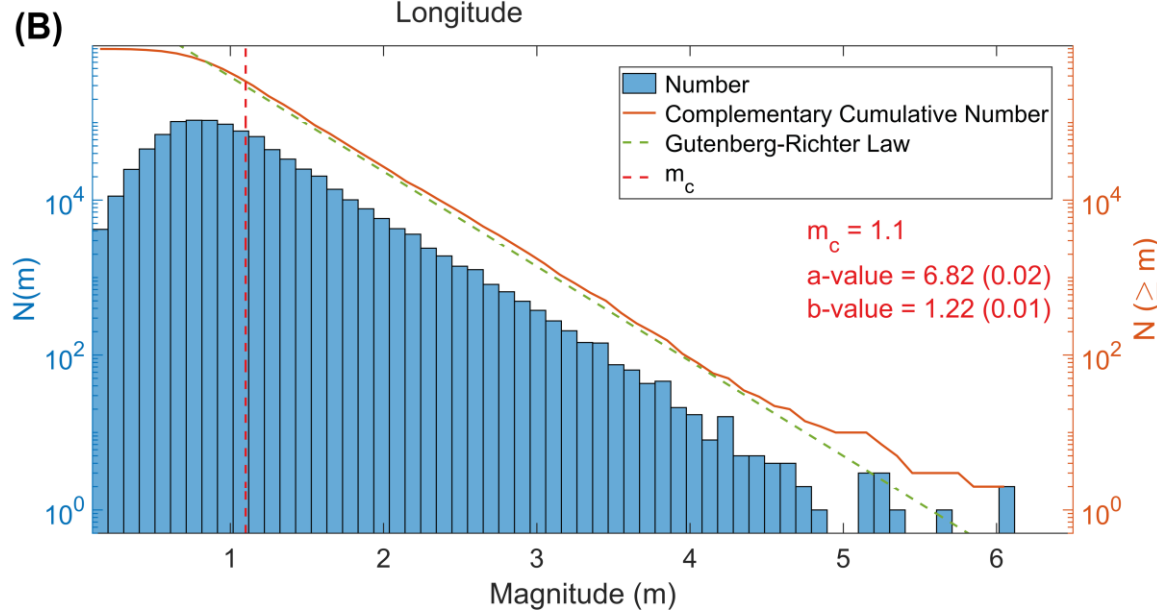
The (apparent) branching ratio, n , decreases with increasing m_0

Are we systematically underestimating n ?

An Upper Bound for m_0 Using AI-enhanced Catalogs



Certainly $m_0 < 1$
Likely $m_0 < 0$



Take home message

A minimum magnitude of triggering, m_0 , is usually set equal to the catalog cut-off magnitude in ETAS calibrations and simulations without testing for it.

- 1) **A new version of the ETAS model with smooth crossover triggering magnitude** enables to quantify the transition from non-triggering to triggering reliably.
- 2) **The crossover magnitude, m_0 , is well below the completeness magnitude** of the best seismic catalogs available today.
- 3) **The arbitrary selection of m_0 as being equal to the completeness magnitude in ETAS models induce biases in the estimation of ETAS parameters.**

Davide Zaccagnino

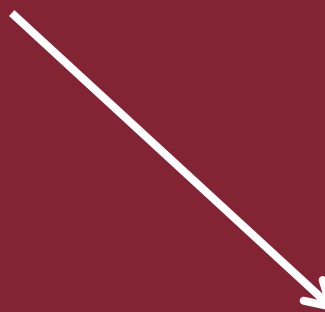
Post-doctoral Researcher

zaccagnino@sustech.edu.cn

www.davidezaccagnino.com

Southern University of Science and Technology (SUSTech)
SUSTech Academy for Advanced Interdisciplinary Studies
Institute of Risk Analysis, Prediction and Management

My webpage



感谢